Correlation and Regression Analysis

Contents

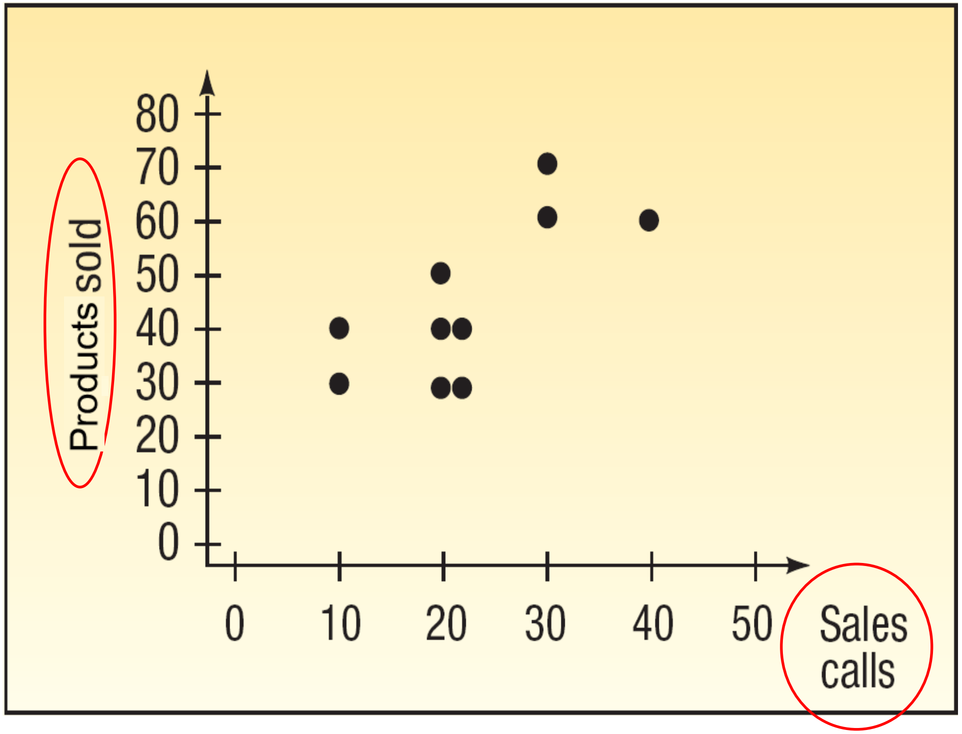
- The **relationship** between **two** variables using **correlation coefficient**.
- **Dependent** and **independent** variable.
- **Regression analysis** to estimate the **linear relationship** between **two** variables
- **Interpretation** of regression analysis.
- Significance of **slope** of the regression equation.
- **Regression equation** to **predict** the **dependent** variable.
- **Coefficient** of **determination**.

Scatter Diagram Example

A sales manager of Chaldal.com wants to determine whether there is a **relationship between** the **number of sales calls made** in a month and the **number of products sold** that month. The manager selects a **random sample of 10 representatives** and determines the **number of sales calls** each representative made last month and the **number of products** sold.

Scatter Diagram Example

Sales Representative	Number of Sales Calls	Number of Products Sold
1	20	30
2	40	60
3	20	40
4	30	60
5	10	30
6	10	40
7	20	40
8	20	50
9	20	30
10	30	70

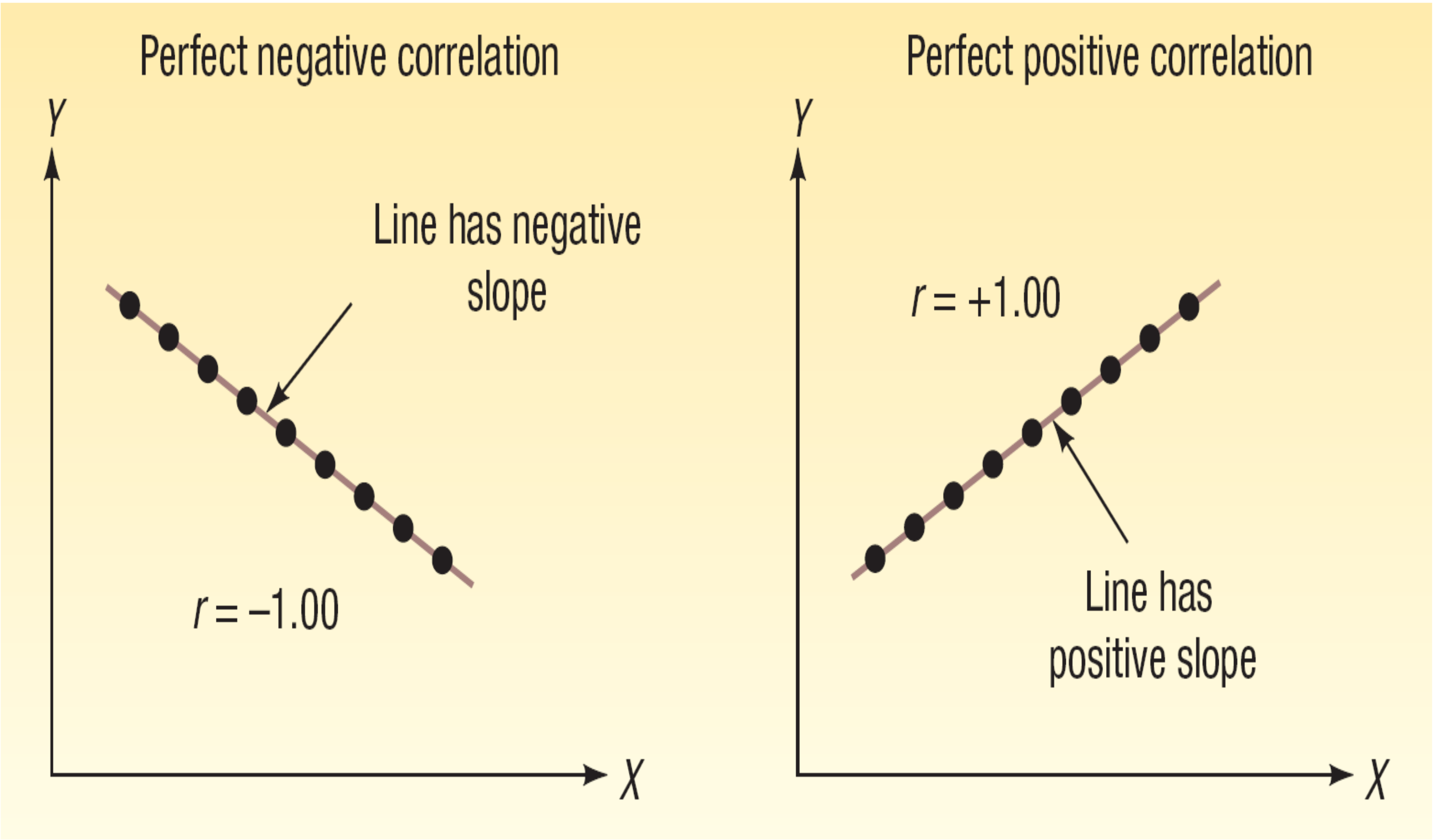


The Coefficient of Correlation (r)

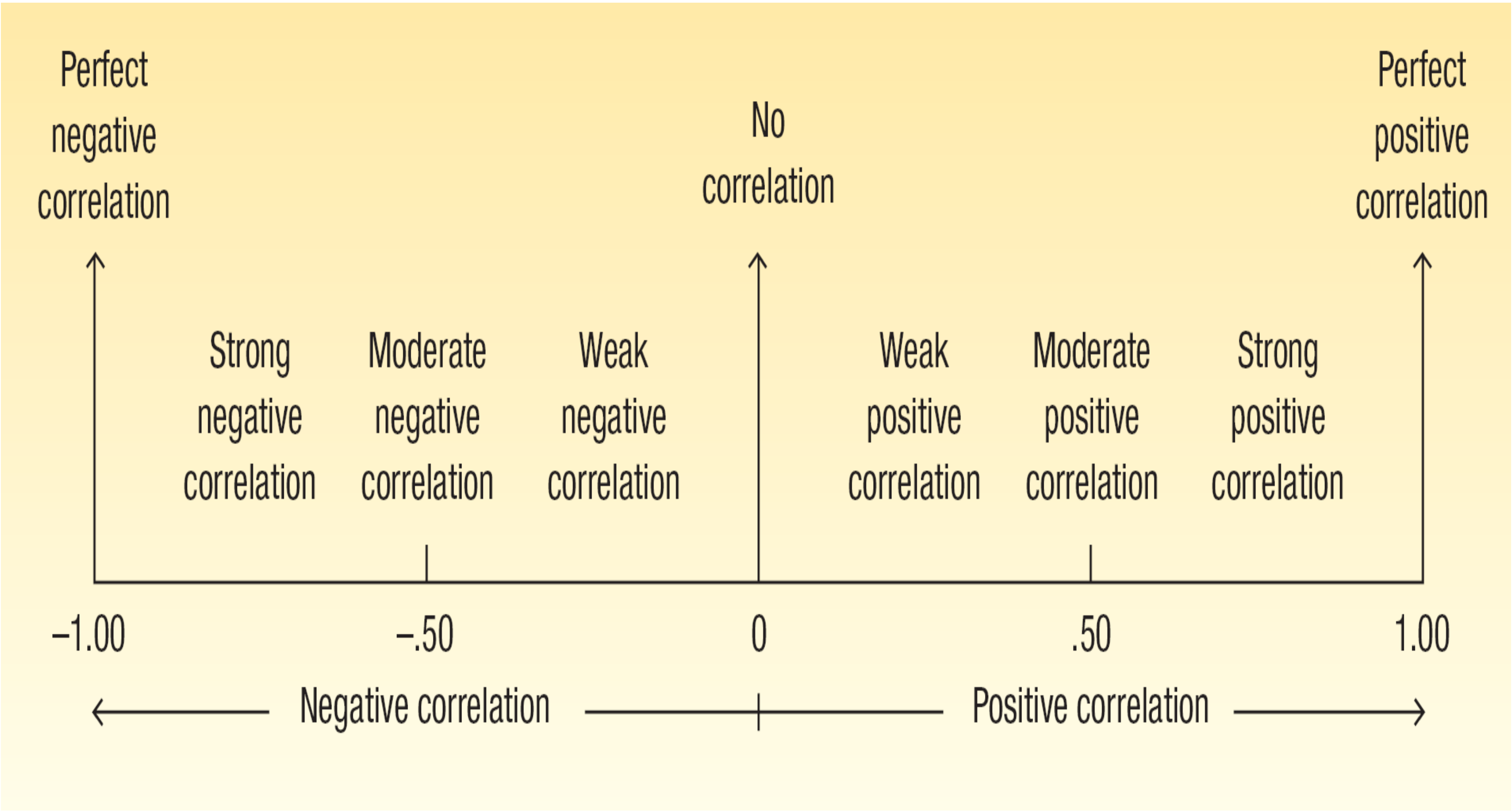
The **Coefficient of Correlation** (*r*) is a measure of the **strength** of the **relationship** between **two variables**.

- It shows the **direction** and **strength** of the linear relationship between two **interval** or **ratio-scale** variables.
- It can **range** from -1.00 to +1.00.
- Values of -1.00 or +1.00 indicate **perfect** and **strong** correlation.
- Values **close to 0** indicate **weak** correlation.
- **Negative** values indicate an **inverse** relationship and **positive** values indicate a **direct** relationship.

The Coefficient of Correlation (r)

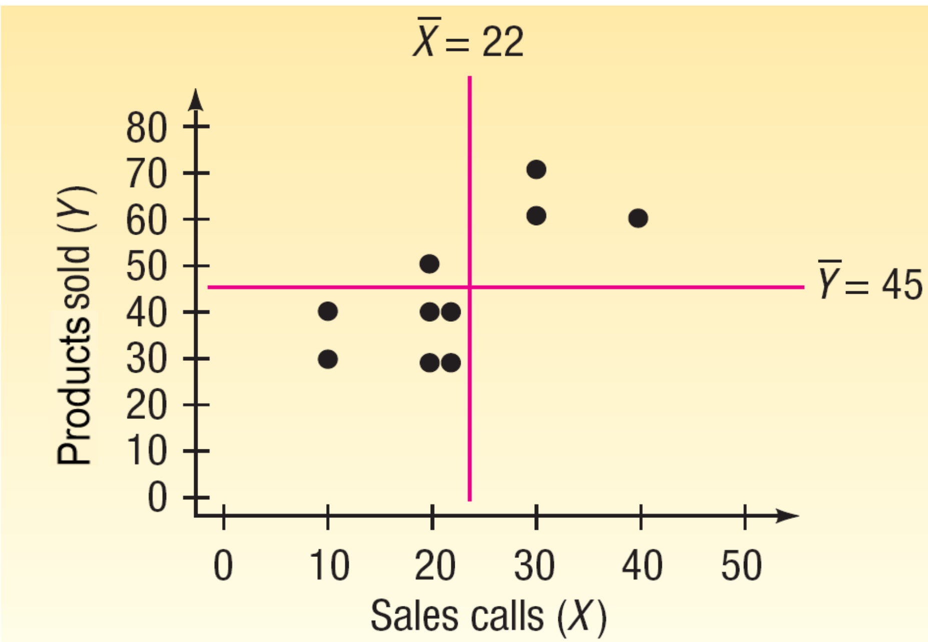


Correlation Coefficient - **Interpretation**



Correlation Coefficient - Example

Using the Copier Sales of America data which a scatterplot is shown below, compute the **correlation coefficient**.



Sales Representative	Number of Sales Calls	Number of Products Sold
1	20	30
2	40	60
3	20	40
4	30	60
5	10	30
6	10	40
7	20	40
8	20	50
9	20	30
10	30	70

Correlation Coefficient - Example

CORRELATION COEFFICIENT

$$r = \frac{\Sigma(X - \bar{X})(Y - \bar{Y})}{(n - 1)s_x s_y}$$

Sales Representative	Sales Calls, X	Products Sold, Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$
1	20	30	−2	−15	30
2	40	60	18	15	270
3	20	40	−2	−5	10
4	30	60	8	15	120
5	10	30	−12	−15	180
6	10	40	−12	−5	60
7	20	40	−2	−5	10
8	20	50	−2	5	−10
9	20	30	−2	−15	30
10	30	70	8	25	200
					900

$$r = \frac{\Sigma(X - \bar{X})(Y - \bar{Y})}{(n - 1)s_x s_y} = \frac{900}{(10 - 1)(9.189)(14.337)} = 0.759$$

Rank-Order Correlation

- **Spearman's** coefficient of rank correlation reports the **association** between **two** sets of **ranked** observations.
- The **features** are:
  - It can range from **–1.00** up to **1.00**.
  - It is similar to **Pearson's** coefficient of correlation but is based on **ranked data**.
  - A measure of correlation for **ordinal-level** data.

Rank-Order Correlation

SPEARMAN'S COEFFICIENT OF RANK CORRELATION

$$r_s = 1 - \frac{6\Sigma d^2}{n(n^2 - 1)}$$

where:

d is the difference between the ranks for each pair.

n is the number of paired observations.

Rank-Order Correlation Example

Bengal Plastics, Inc., **recruits management trainees** at colleges and universities throughout Bnagladesh. **Each trainee** is given a **score**, an **expression of future potential** ranging from **0 to 200**, by the recruiter during the on-campus interview. A **higher score** indicates **more potential**. An applicant hired by Bengal then enters an in-plant training program. At the completion of this program the recruit is given **another composite score** (**0 to 100**), which is based on tests, opinions of group leaders, and in-plant training officers. A **higher score** indicates **more potential.** Is there an **association** between the **on-campus and in-plant** scores?

Rank-Order Correlation Example

The **on-campus scores** and the **in-plant training scores** are on the table below:

Graduate	On-Campus Score	In-Plant Score
Spina, Sal	83	45
Gordon, Ray	106	45
Althoff, Roberta	92	45
Alvear, Ginny	48	36
Wallace, Ann	127	68
Lyons, George	113	83
Harbin, Joe	118	88
Davison, Jack	78	61
Brydon, Tom	83	66
Bobko, Jack	193	94
Koppel, Marty	101	56
Nyland, Patricia	123	91

Rank-Order Correlation Example

Calculations for the Coefficient of Rank Correlation (r_s)

	Scores		Rank		Difference between	
Graduate	On-Campus Score	In-Plant Score	On-Campus Rank	In-Plant Rank	Ranks (d)	d^2
Spina, Sal	83	45				
Gordon, Ray	106	45				
Debbisi, Ray	92	45				
Alvear, Ginny	48	36				
Wallace, Ann	127	68				
Lyons, George	113	83				
Harbin, Joe	118	88				
Davison, Jack	78	61				
Brydon, Tom	83	66				
Bobko, Jack	193	94				
Koppel, Marty	101	56				
Nyland, Patricia	123	91				

Rank-Order Correlation Example

Calculations for the Coefficient of Rank Correlation (r_s)

	Scores		Rank		Difference between	
Graduate	On-Campus Score	In-Plant Score	On-Campus Rank	In-Plant Rank	Ranks (d)	d^2
Spina, Sal	83	45	3.5	3		
Gordon, Ray	106	45	7	3		
Debbisi, Ray	92	45	5	3		
Alvear, Ginny	48	36	1	1		
Wallace, Ann	127	68	11	8		
Lyons, George	113	83	8	9		
Harbin, Joe	118	88	9	10		
Davison, Jack	78	61	2	6		
Brydon, Tom	83	66	3.5	7		
Bobko, Jack	193	94	12	12		
Koppel, Marty	101	56	6	5		
Nyland, Patricia	123	91	10	11		

Rank-Order Correlation Example

Calculations for the Coefficient of Rank Correlation (r_s)

	Scores		Rank		Difference between	
Graduate	On-Campus Score	In-Plant Score	On-Campus Rank	In-Plant Rank	Ranks (d)	d^2
Spina, Sal	83	45	3.5	3	0.5	0.25
Gordon, Ray	106	45	7	3	4.0	16.00
Debbisi, Ray	92	45	5	3	2.0	4.00
Alvear, Ginny	48	36	1	1	0.0	0.00
Wallace, Ann	127	68	11	8	3.0	9.00
Lyons, George	113	83	8	9	-1.0	1.00
Harbin, Joe	118	88	9	10	-1.0	1.00
Davison, Jack	78	61	2	6	-4.0	16.00
Brydon, Tom	83	66	3.5	7	-3.5	12.25
Bobko, Jack	193	94	12	12	0.0	0.00
Koppel, Marty	101	56	6	5	1.0	1.00
Nyland, Patricia	123	91	10	11	-1.0	1.00
					0	**61.50**

Rank-Order Correlation Example

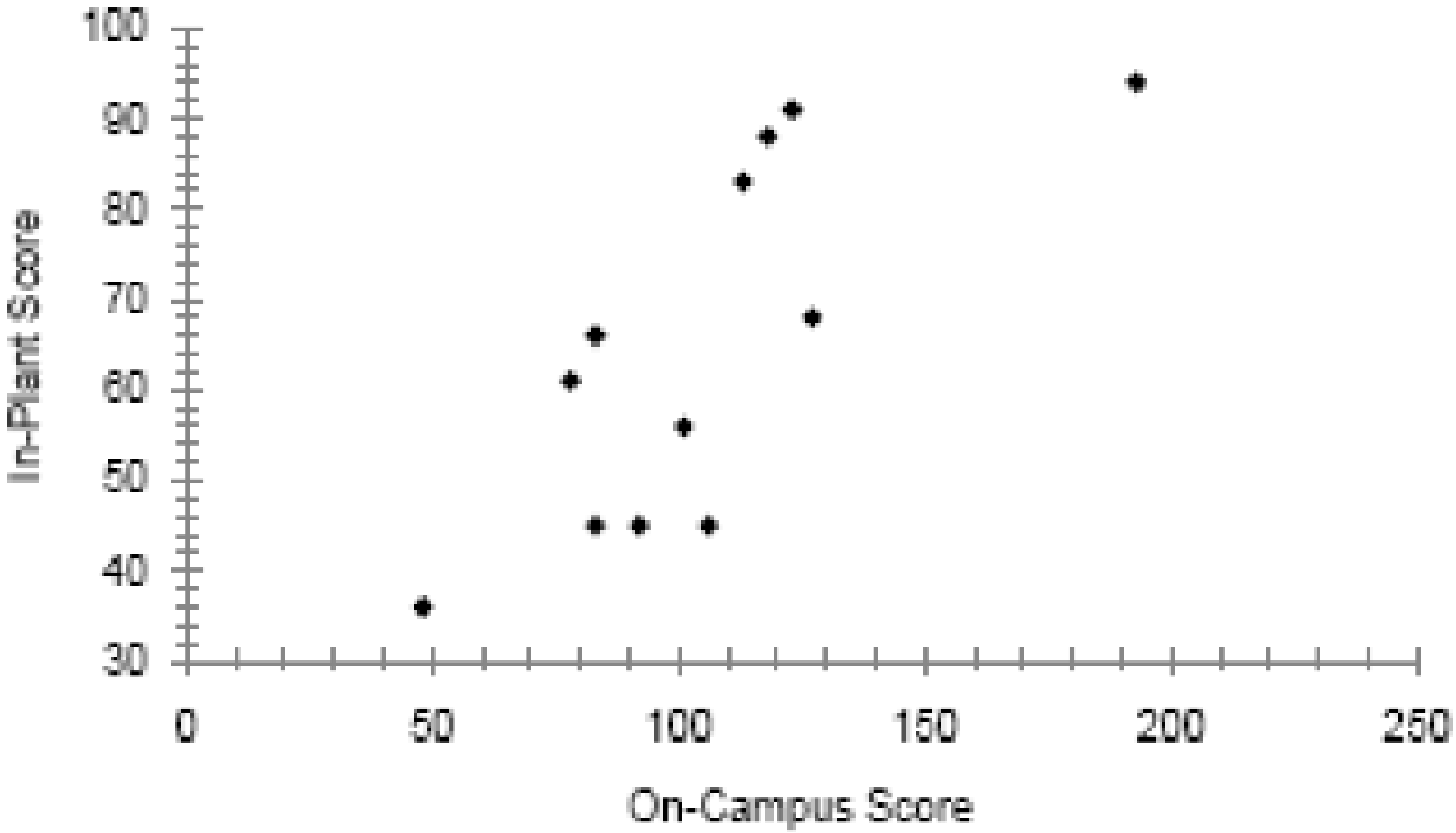
$$r_s = 1 - \frac{6\sum d^2}{n(n^2-1)} = 1 - \frac{6(61.50)}{12(12^2-1)} = 1 - .215 = .785$$

Conclusion:

The value of 0.785 indicates a **strong positive** association between the **ratings** of the on-campus recruiter and the scores of the in-plant training staff.

The graduates that received **high ratings** from the **on-campus recruiter** also tended to be the ones that received **high ratings** from the **training staff**.

Rank-Order Correlation Example



Correlation Coefficient - Example

What does correlation of 0.759 mean?

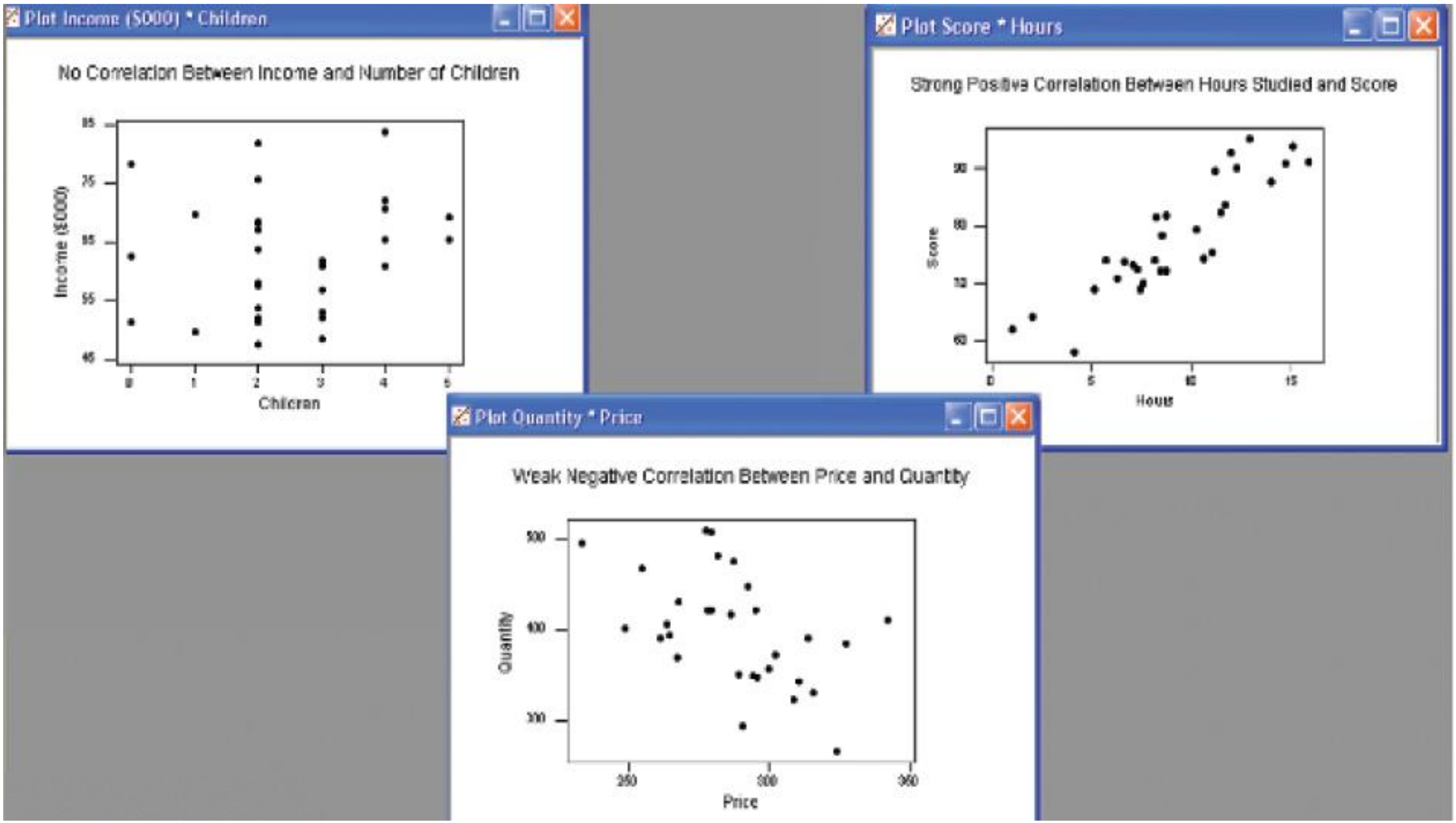
First, it is **positive**, so we see there is a **direct relationship** between the number of **sales calls** and the number of p**roducts sold**.

The value of 0.759 is **fairly close to 1.00**, so we conclude that the association is **strong**.

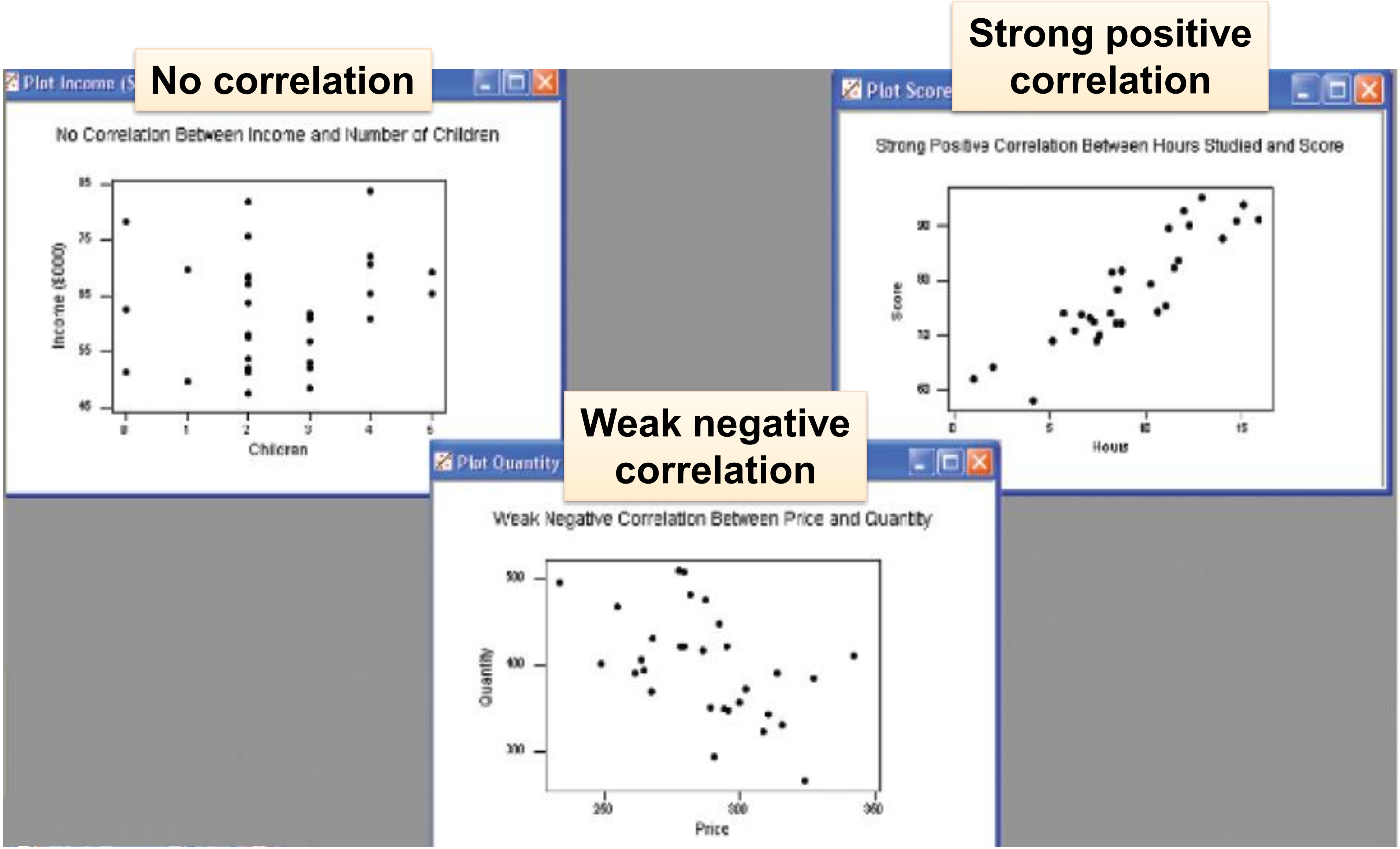
*However, does this mean that **more** sales calls **cause more** sales?*

No, we have **not** demonstrated **cause** and **effect** here, only that the two variables—sales calls and products sold—are **related**.

Scatter Plots



Scatter Plots



Regression Analysis Introduction

- Numerical measures to express the **strength of relationship** between two variables are developed.
- In addition, an **equation** is used to express the relationship between variables, allowing us to **estimate** one variable on the basis of another.

Regression Analysis Introduction

EXAMPLES

- Does the **amount** Square Group spends per month on **training** its sales force affect its **monthly sales**?
- Is the **number of square feet** in a home related to the **cost to heat** the home in January?
- In a study of fuel efficiency, is there a relationship between **miles per gallon** and the **weight of a car**?
- Does the **number of hours** that students studied for an exam influence the **exam score**?

Dependent vs. Independent Variable

The Dependent **Variable** is the **variable** being **predicted** or **estimated**.

The Independent **Variable** provides the **basis** for **estimation**. It is the **predictor** variable.

Which in the questions below are the **dependent** and **independent** variables?

i. Does the **amount** Square Group spends per month on **training** its sales force affect its **monthly sales**?
ii. Is the **number of square feet** in a home related to the **cost to heat** the home in January?
iii. Does the **number of hours** that students studied for an exam influence the **exam score**?

Regression: **Terminology**

Dependent variable	Independent variable
⇕	⇕
Explained variable	**Explanatory variable**
⇕	⇕
Predictand	Predictor
⇕	⇕
Regressand	**Regressor**
⇕	⇕
Response	Stimulus
⇕	⇕
Endogenous	Exogenous
⇕	⇕
Outcome	Covariate
⇕	⇕
Controlled variable	Control variable

Regression Analysis

In regression analysis we use the **independent** variable (*X*) to estimate the **dependent** variable (*Y*).

- The relationship between the variables is **linear**.
- Both variables must be **at least interval scale**.
- The **least squares criterion** is used to determine the equation.

Regression Equation An equation that expresses the **linear relationship** between two variables.

Least Squares Principle Determining a regression equation by **minimizing** the **sum** of the **squares** of the **vertical distances** between the **actual *Y* values** *and* the **predicted values of *Y*.**

Linear Regression Model

GENERAL FORM OF LINEAR REGRESSION EQUATION $\hat{Y} = a + bX$

where

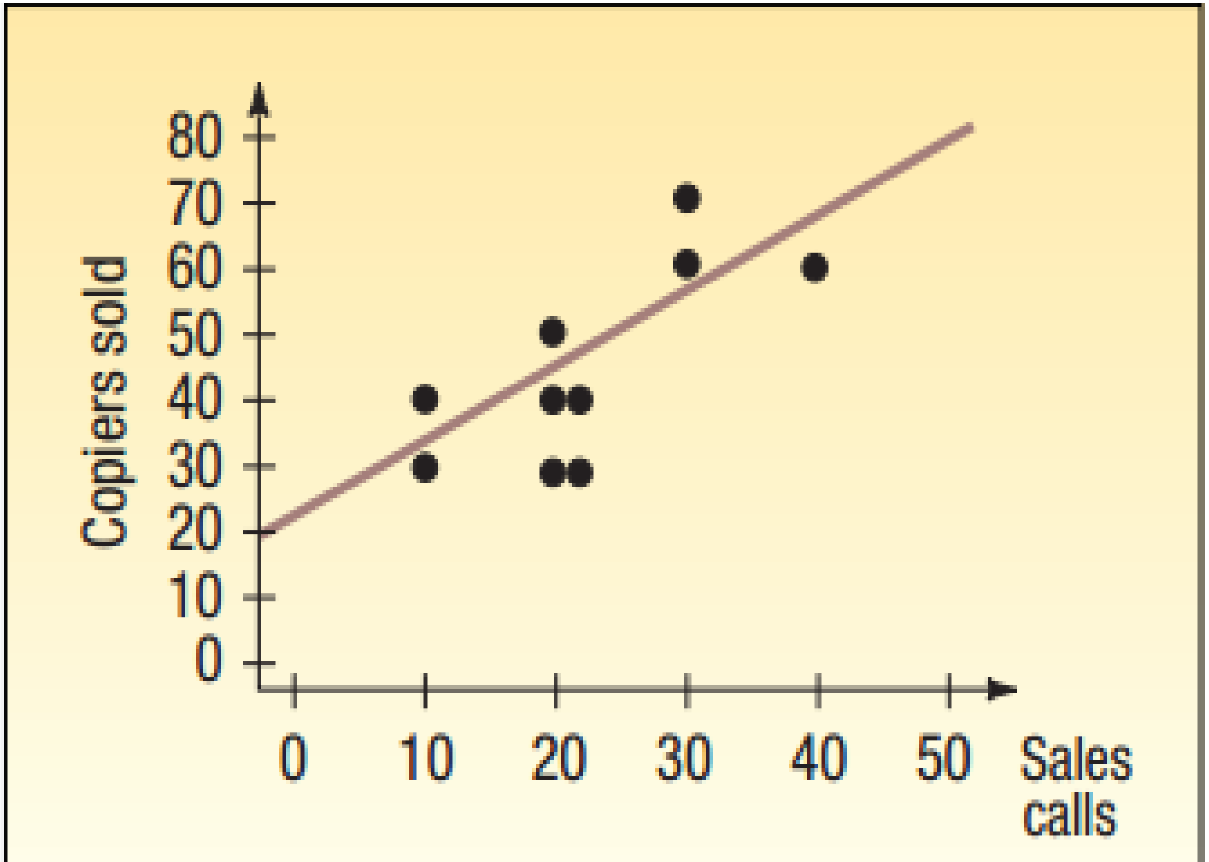
$\hat{Y}$ read Y hat, is the estimated value of the Y variable for a selected X value.

a is the Y-intercept. It is the estimated value of Y when $X = 0$. Another way to put it is: a is the estimated value of Y where the regression line crosses the Y-axis when X is zero.

b is the slope of the line, or the average change in $\hat{Y}$ for each change of one unit (either increase or decrease) in the independent variable X.

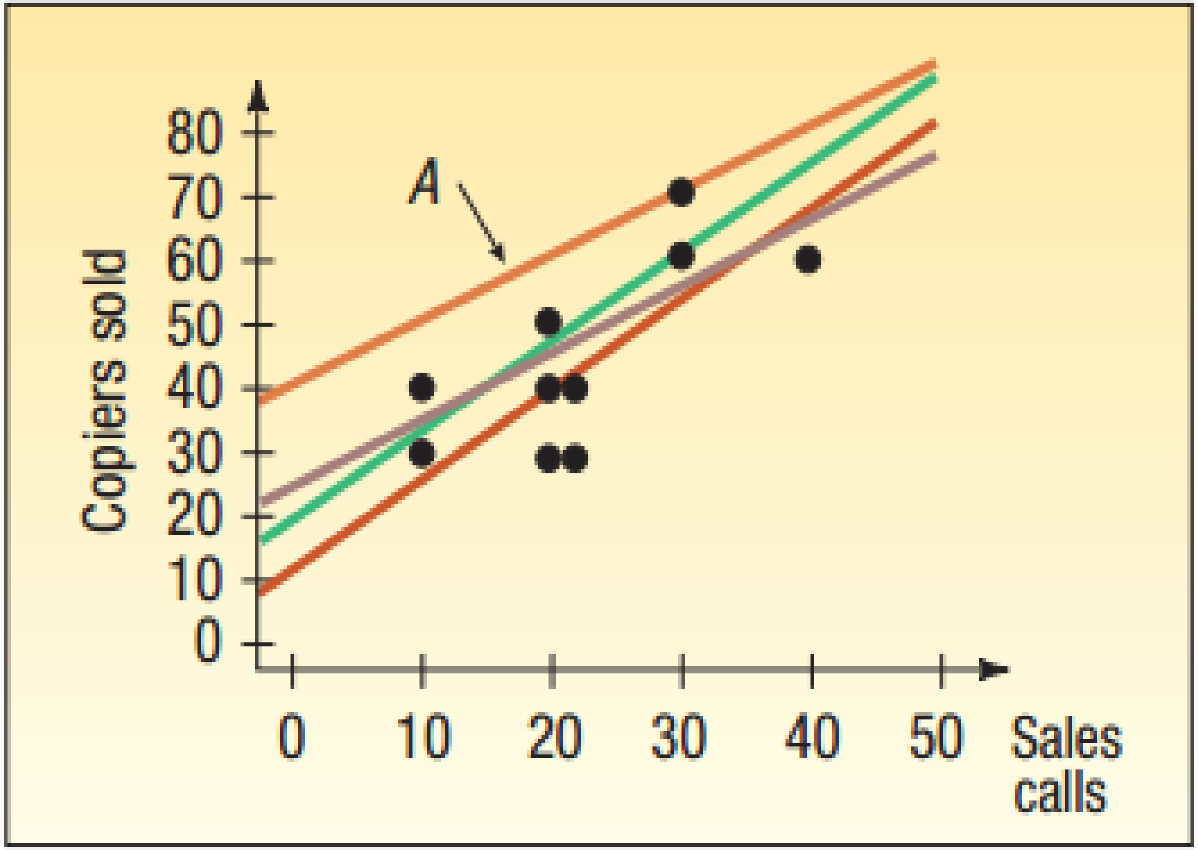
X is any value of the independent variable that is selected.

Regression Analysis – Least Squares Principle



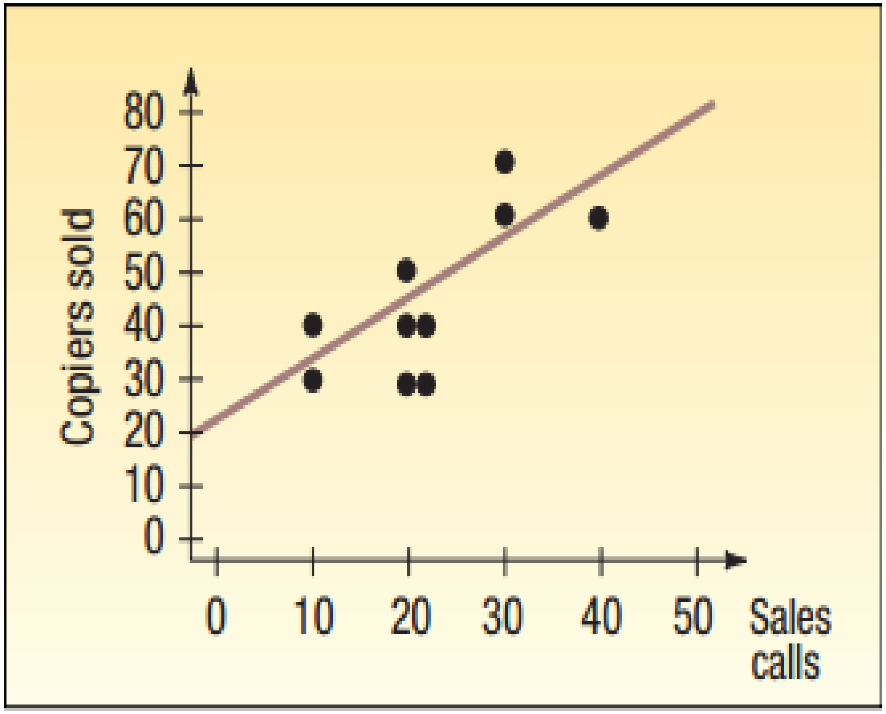
Sales Calls and Copiers Sold for 10 Sales Representatives

Regression Analysis – Least Squares Principle

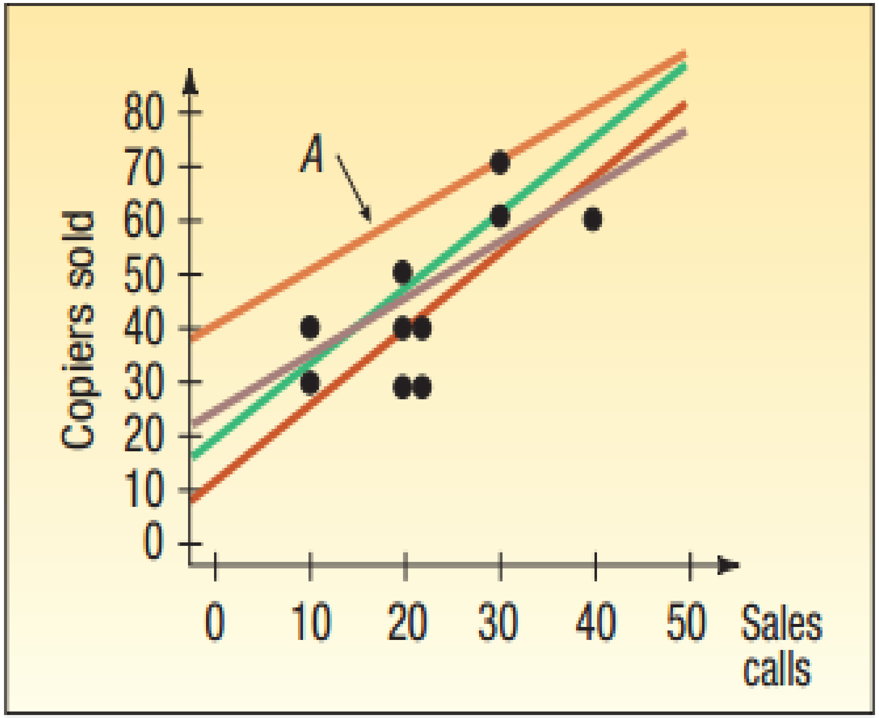


Four Lines Superimposed on the Scatter Diagram

Regression Analysis – Least Squares Principle



Sales Calls and Copiers Sold for 10 Sales Representatives



Four Lines Superimposed on the Scatter Diagram

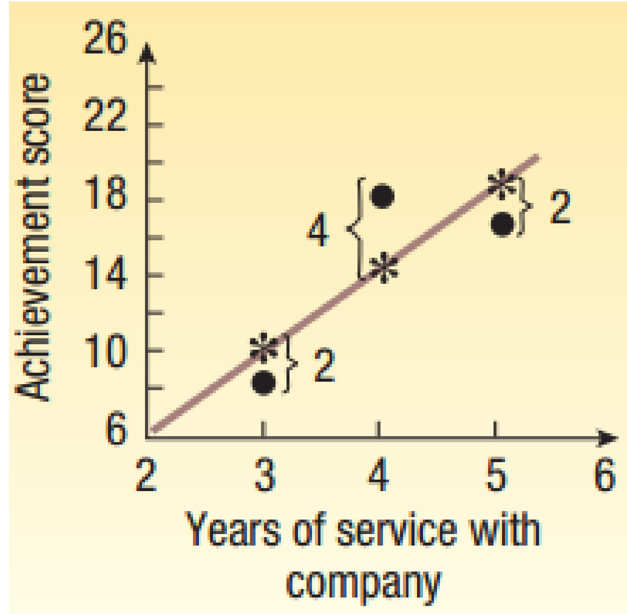
Regression Analysis – Least Squares Principle

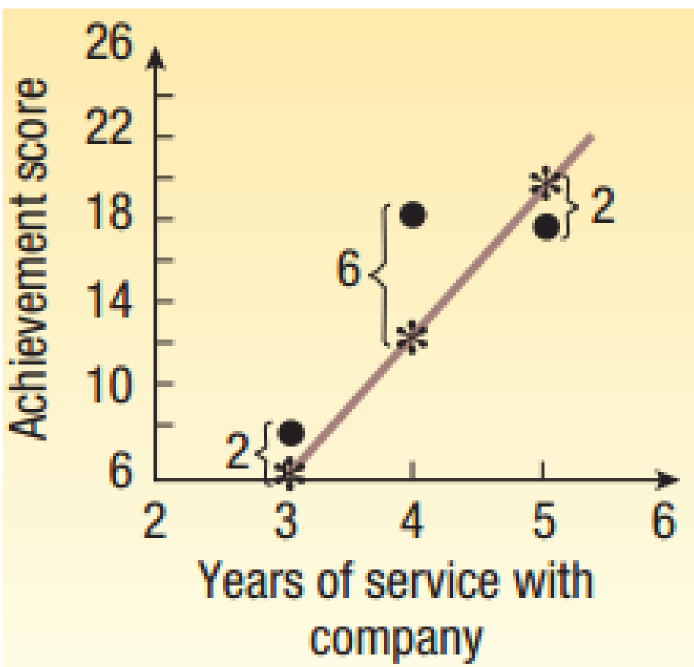
- **LEAST SQUARES PRINCIPLE**

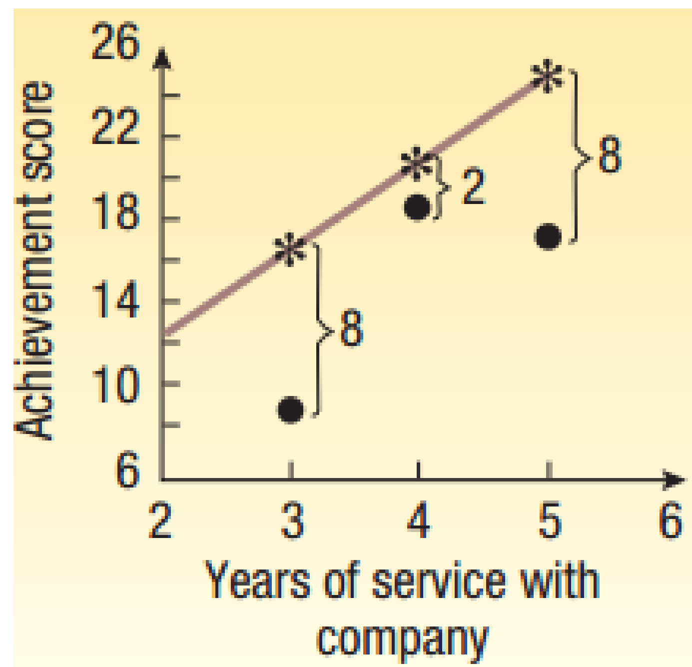
  A mathematical **procedure** that uses the data to position a line with the objective of **minimizing** the **sum of** the **squares** of the **vertical distances** between the **actual Y** values and the **predicted** values of **Y**.

Regression Analysis

Least Squares Principle







Computing Slope and Y-intercept

SLOPE OF THE REGRESSION LINE

$$b = r\frac{s_y}{s_x}$$

where

r is the correlation coefficient.

s_y is the standard deviation of Y (the dependent variable).

s_x is the standard deviation of X (the independent variable).

***Y*-INTERCEPT**

$$a = \overline{Y} - b\overline{X}$$

where

$\overline{Y}$ is the mean of Y (the dependent variable).

$\overline{X}$ is the mean of X (the independent variable).

Regression Equation - Example

The sales manager gathered information on the number of sales calls made and the number of products sold for a random **sample of 10** sales representatives. Use the **least squares method** to **determine** a **linear equation** to express the relationship between the two variables.

What is the **expected number of products sold** by a representative who **made 20 calls**?

Sales Representative	Number of Sales Calls	Number of Products Sold
1	20	30
2	40	60
3	20	40
4	30	60
5	10	30
6	10	40
7	20	40
8	20	50
9	20	30
10	30	70

Finding and Fitting the Regression Equation - Example

Step 1 – Find the **slope** (*b*) of the line

$$b = r\left(\frac{s_y}{s_x}\right) = .759\left(\frac{14.337}{9.189}\right) = 1.1842$$

Step 2 – Find the ***y*-intercept** (*a*)

$$a = \overline{Y} - b\overline{X} = 45 - 1.1842(22) = 18.9476$$

Regression equation:

$\hat{Y}$ = a + bX

$\hat{Y}$ = 18.9476 + 1.1842 X

$\hat{Y}$ = 18.9476 + 1.1842 (20)

$\hat{Y}$ = 42.6316

Testing Significance of Slope – Copier Sales Example

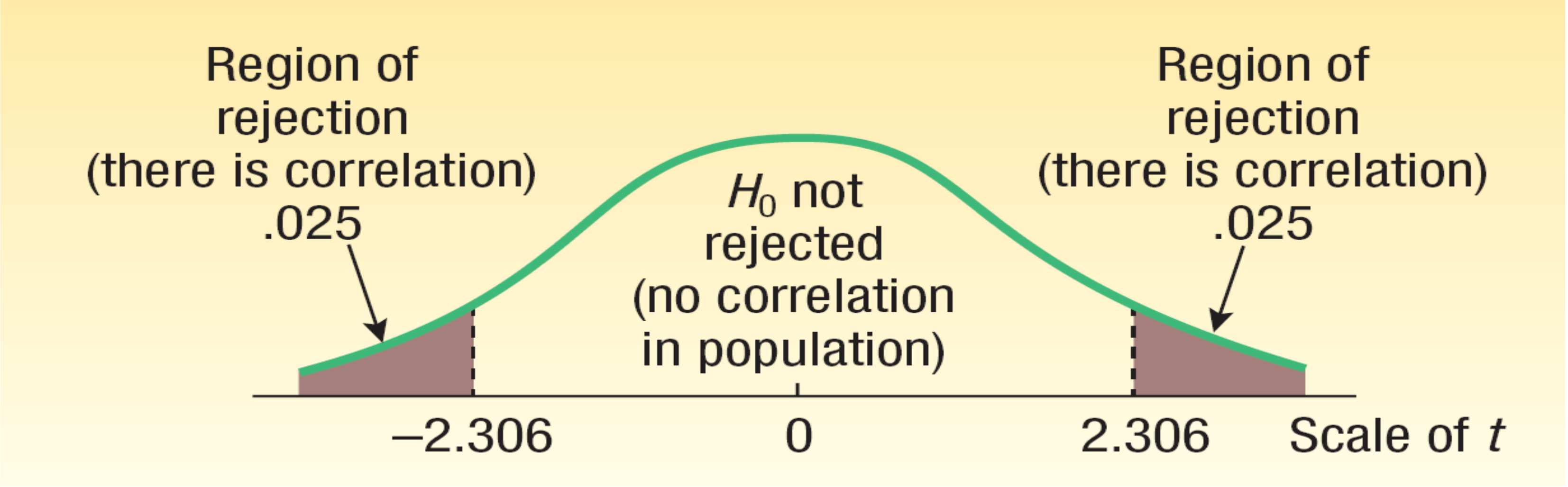
H_0: β = 0 (the slope of the linear model is 0)

H_1: β ≠ 0 (the slope of the linear model is not 0)

Reject H_0 if: $t > t_{\alpha/2,n-2}$ or $t < -t_{\alpha/2,n-2}$

$t > t_{0.025,8}$ or $t < -t_{0.025,8}$

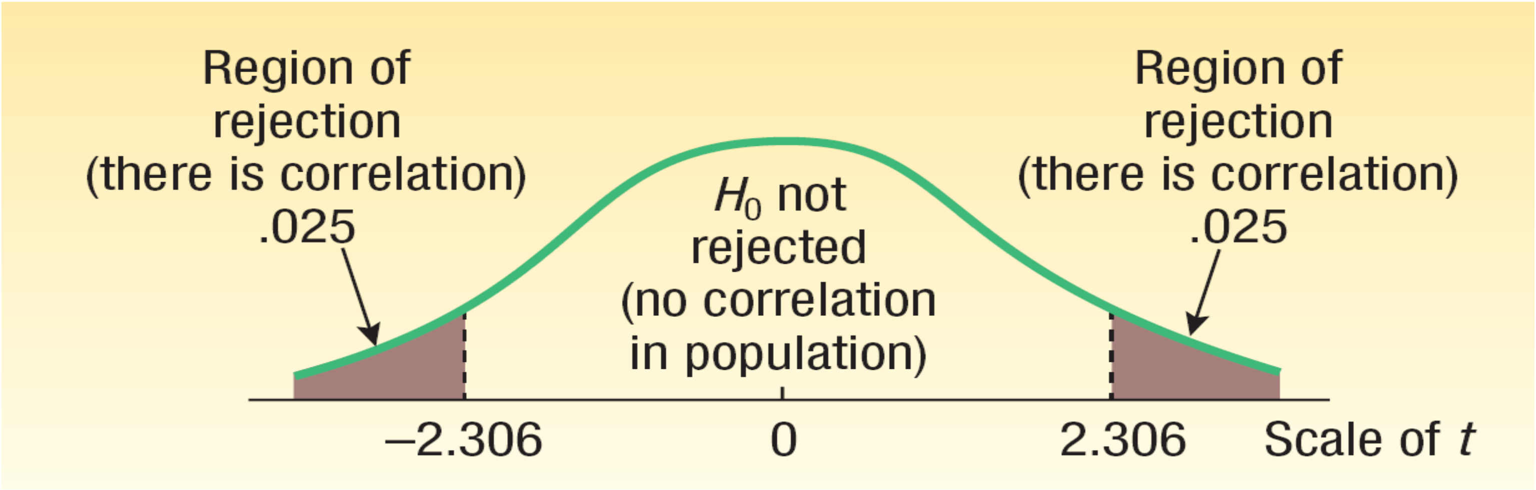
$t > 2.306$ or $t < -2.306$



Testing Significance of Slope – Copier Sales Example

Compute t statistic and make a conclusion:

$$t = \frac{b - 0}{s_b} = \frac{1.1842 - 0}{0.3591} = 3.297$$



Conclusion: The **slope** of the equation is **significantly** different from zero.

Testing Significance of Slope – Product Sales Example

Calls	Sales
20	30
40	60
20	40
30	60
10	30
10	40
20	40
20	50
20	30
30	70

SUMMARY OUTPUT

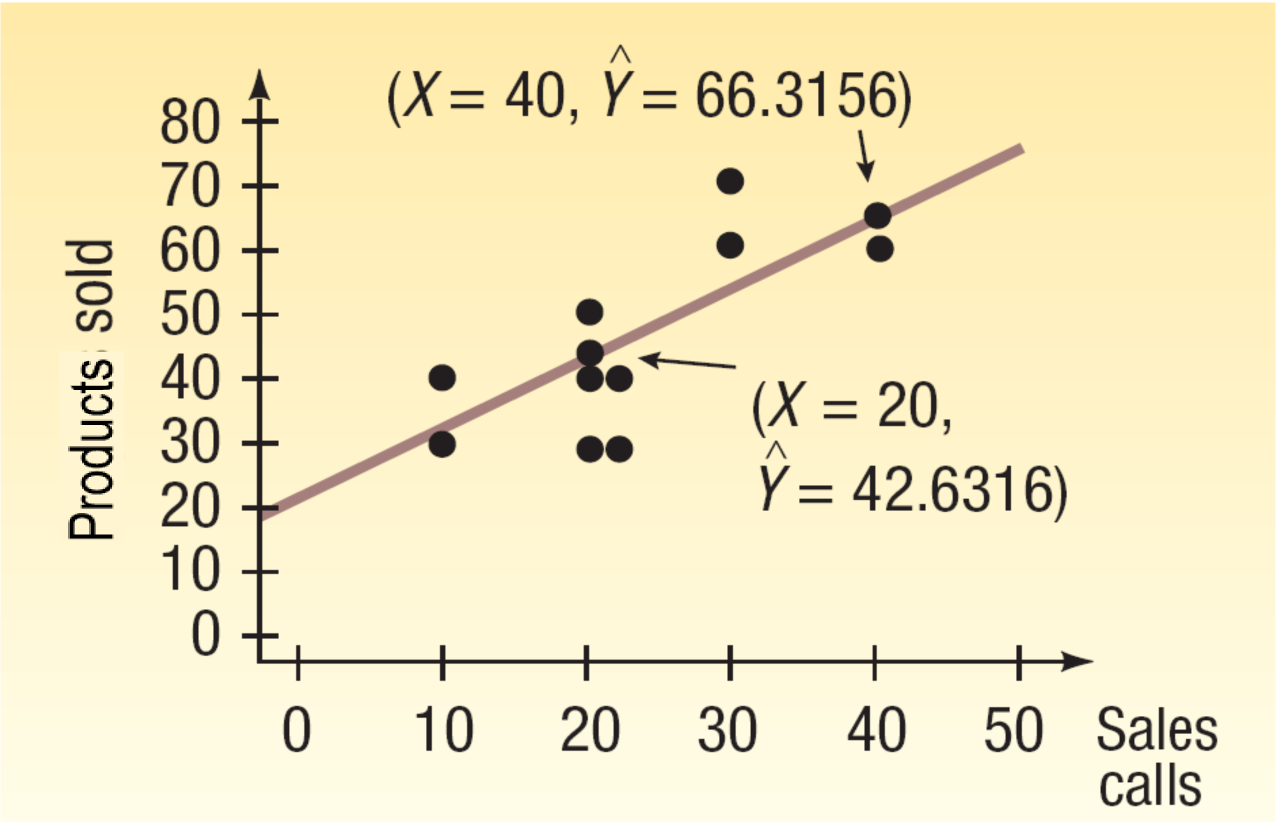
Regression Statistics	
Multiple R	0.759
R Square	0.576
Adjusted R Square	0.523
Standard Error	9.901
Observations	10

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	1065.789	1065.789	10.872	0.011
Residual	8	784.211	98.026		
Total	9	1850.000			

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	18.9474	8.4988	2.2294	0.05635
Calls	1.18421	0.35914	3.29734	0.01090

Plotting Estimated and Actual Y's



Coefficient of Determination (r^2)

The **coefficient of determination** (r^2) is the **proportion** of the **total variation** in the **dependent variable** (Y) that is **explained** by the **variation** in the **independent variable** (X).

- It is the **square** of the **coefficient of correlation**.
- It **ranges** from **0 to 1**.
- It does **not** give any information on the **direction** of the relationship between the variables.

Coefficient of Determination (r^2) – Product Sales Example

- The **coefficient of determination**, r^2 ,is 0.576, found by $(0.759)^2$
- This is a **proportion** or a **percent**; we can say that **57.6 percent** of the **variation** in the **number of products sold is explained** by the **variation** in the **number of sales calls**.

Learning Objectives (LO)

LO1 Calculate, test, and interpret the relationship between two variables using correlation coefficient.

LO2 Define dependent and independent variable.

LO3 Apply regression analysis to estimate the linear relationship between two variables

LO4 Interpret the regression analysis.

LO5 Evaluate the significance of the slope of the regression equation.

LO6 Calculate and interpret coefficient of determination.

Multiple Linear Regression - Example

BD Real Estate Ltd. sells homes along the northern part of Bangladesh. One of the questions most frequently asked by prospective buyers is: If we purchase this home, how much can we expect to **pay to heat** it during the **winter**? The research department at BD Real Estate has been asked to develop some guidelines regarding **heating costs** for single-family homes.

Three variables are thought to be related to the heating costs: (1) the mean daily **outside temperature**, (2) the **number** of **inches** of **insulation** in the attic, and (3) the **age** in years of the **furnace**.

Multiple Linear Regression - Example

To investigate, the research department selected a **random sample of 20** recently sold homes. It determined the **cost to heat** each home last **January**, as well as the January **outside temperature** in the region, the number of inches of **insulation** in the attic, and the **age** of the furnace.

Multiple Linear Regression - Example

Home	Heating Cost BDT	Mean Outside Temperature (°F)	Attic Insulation (inches)	Age of Furnace (years)
1	250	35	3	6
2	360	29	4	10
3	165	36	7	3
4	43	60	6	9
5	92	65	5	6
6	200	30	5	5
7	355	10	6	7
8	290	7	10	10
9	230	21	9	11
10	120	55	2	5
11	73	54	12	4
12	205	48	5	1
13	400	20	5	15
14	320	39	4	7
15	72	60	8	6
16	272	20	5	8
17	94	58	7	3
18	190	40	8	11
19	235	27	9	8
20	139	30	7	5

Multiple Linear Regression - Example

Home	Y Heating Cost BDT	X_1 Mean Outside Temperature (°F)	X_2 Attic Insulation (inches)	X_3 Age of Furnace (years)
1	250	35	3	6
2	360	29	4	10
3	165	36	7	3
4	43	60	6	9
5	92	65	5	6
6	200	30	5	5
7	355	10	6	7
8	290	7	10	10
9	230	21	9	11
10	120	55	2	5
11	73	54	12	4
12	205	48	5	1
13	400	20	5	15
14	320	39	4	7
15	72	60	8	6
16	272	20	5	8
17	94	58	7	3
18	190	40	8	11
19	235	27	9	8
20	139	30	7	5

Multiple Linear Regression – Excel Output

SUMMARY OUTPUT	
Regression Statistics	
Multiple R	0.8968
R Square	0.8042
Adjusted R Square	0.7675
Standard Error	51.0486
Observations	20

Multiple Linear Regression – Excel Output

ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	3	171220.50	57073.490	21.901	6.56E-06	
Residual	16	41695.28	2605.955			
Total	19	212915.80				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept (a)	427.194	59.601	7.168	2.24E-06	300.844	553.543
Temp (b_1)	- 4.582	0.772	-5.933	2.1E-05	-6.220	-2.945
Insul (b_2)	-14.831	4.754	-3.119	0.007	-24.910	-4.752
Age (b_3)	6.101	4.012	1.521	0.148	-2.404	14.606

Estimating Multiple Regression Equation

$$\hat{Y} = 427.194 - 4.583X_1 - 14.831X_2 + 6.101X_3$$

Interpreting the Regression Coefficients:

The regression coefficient for mean **outside temperature** (X_1) is - 4.583.

- For every unit increase in **temperature**, holding the other two independent variables constant, monthly **heating cost** is expected to decrease by $4.583.

The **attic insulation** variable, X_2, also shows a **negative** relationship.

- For each additional inch of **insulation**, the **cost to heat** the home is expected to decline by $14.83 per month.

Estimating the Multiple Regression Equation

$$\hat{Y} = 427.194 - 4.583X_1 - 14.831X_2 + 6.101X_3$$

Interpreting the Regression Coefficients

The **age of** the **furnace** variable shows a **positive** relationship.

- For each additional year older the **furnace** is, the **cost** is expected to increase by $6.10 per month.

Using the Multiple Regression Equation

Applying the Model for Estimation

What is the **estimated heating cost** for a home if the mean outside temperature is **30 degrees**, there are **5 inches** of insulation in the attic, and the furnace is **10 years** old?

Using the Multiple Regression Equation

<u>Applying the Model for Estimation</u>

What is the **estimated heating cost** for a home if the mean outside temperature is **30 degrees**, there are **5 inches** of insulation in the attic, and the furnace is **10 years** old?

$$\hat{Y} = 427.194 - 4.583X_1 - 14.831X_2 + 6.101X_3$$

$$\hat{Y} = 427.194 - 4.583(30) - 14.831(5) + 6.101(10)$$

$$\hat{Y} = 276.56$$